

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Mechanical Engineering

Comment [U1]: Please enter the name in Arial 12 font

2. **Subject Name:** ThermoFluids Laboratory **L-T-P:** -0-0-3
Credits: 2

Comment [U2]: Please enter the details (use Arial 12 font)

3. **Pre-requisites:** None

Comment [U3]: Please enter the pre-requisites (subject name and subject no. in Arial 12 font)

4. **Syllabus and reference books:**

Comment [U4]: 1. Please type the syllabus and reference books in the box provided below.

Syllabus:

Involves experiments on hydraulics, gas dynamics and heat transfer.

Hydraulics: Characteristics of a centrifugal pump, characteristics of a impulse turbine, study and calibration of flow meters, calibration of an orifice plate by free jet method

Gas Dynamics: Velocity profile in boundary layer flow over a flat plate, pressure distribution over a bluff body for a range of Reynolds numbers, compressible flow through a convergent divergent nozzle, operating characteristics of a centrifugal blower

Heat Transfer: Heat Transfer from a pin fin, forced convective heat transfer in a uniformly heated circular tube, natural convection heat transfer from a heated vertical cylinder, measurement of emissivity

Please use "Arial" font with a font size of "12" and a line spacing of 1.0 for syllabus and reference books.
Please use Justified Text (Ctrl+J) for the content entered in the boxes
Reference books should be numbered as 1), 2), 3)
Please provide all bibliographic details for the reference books

Reference Books:

Not applicable

5. **Lecture-wise break-up:** Not applicable as this is a laboratory course

Comment [U5]: Please provide lecture-wise break-up in the table given below:

Please use "Arial" font with a font size of "12" and a line spacing of 1.0

Comment [U6]:

Comment [U7]: Please convert the template into PDF and upload it in ERP

Sl. No.	Topic	No. of lectures
1.		
2.		
3.		
4.		
...		
Total number of hours		